

NURSEFLOW

Enterprise Healthcare Scheduling, Gemma 4 AI Patient Simulator & Break Lounge Engine

Document Parameter	Metadata Specification
System Version	NurseFlow v1.0.0 Enterprise Release
Target Audience	Hospital Administrators, Lead Nurses, System Architects & Hackathon Judges
Primary Tech Stack	React 18 (Vite), Node.js, Express, Prisma ORM, SQLite, Socket.IO, Google Gemma 4 AI
Repository URL	https://github.com/Sagheer1122/Arbisoft-GDG-Hackathon.git
Documentation Status	Complete Master System & API Manual (10–15 Pages Detailed Specifications)

Notice of Confidentiality: This document contains proprietary system architecture specifications, database schemas, and AI algorithm designs for the NurseFlow platform. Authorized for internal hospital operations and hackathon review.

1. Executive Summary & Industry Problem Analysis

Modern hospital environments across Pakistan and globally face severe operational and human resource challenges. Nursing personnel work under extreme stress during 12-hour duty shifts, dealing with complex patient handovers, manual scheduling conflicts, and demanding family interactions.

Core Industry Challenges Addressed:

1. **Scheduling Friction & Overlap:** Manual paper schedules or basic spreadsheets lead to shift collisions, missed duties, and unreadable shift handover notes.
2. **Clinical Nurse Burnout:** Extended duty hours without structured mental resets degrade decision-making confidence and increase clinical turnover rates.
3. **Communication Skill Gaps:** Nurses rarely have a safe environment to practice high-stress patient/family conversations (handling anxious surgical patients or angry relatives).
4. **Delayed Ward Emergency Broadcasts:** Traditional phone calls or manual alerts delay response times during urgent hospital ward code calls.

The NurseFlow Strategic Solution:

NurseFlow delivers a unified glassmorphic platform combining automated duty roster intelligence, Google Gemma 4 AI clinical patient roleplay simulation, an AI-powered 10-minute shift break relaxation lounge, and real-time Socket.IO emergency alerts.

2. Full-Stack System Architecture & Technology Ecosystem

NurseFlow follows a decoupled client-server architecture. The frontend application operates as a Single-Page Application (SPA) communicating with the Node.js Express backend via REST APIs and Socket.IO WebSockets.

Architecture Layer	Technologies & Libraries	Responsibility & Architectural Purpose
Frontend Client (SPA)	React 18, Vite, TypeScript, Tailwind CSS	Renders responsive user interfaces with glassmorphic UI design tokens.
State & Security Context	React Context API, JWT Storage	Global AuthContext, SocketContext, and protected route access guards.
Backend REST API	Node.js, Express, TypeScript	Handles REST endpoints, JWT authentication, and request middleware.
Database & ORM	Prisma ORM, SQLite (dev.db)	Type-safe query generation, schema migrations, and relational integrity.
AI Intelligence Engine	Google Gemma 4 AI Engine	Roleplay persona generation, 8-score analysis, and Minimax Game AI.
Real-Time Messaging	Socket.IO (WebSockets)	Instant bi-directional emergency broadcasts and unread alert counters.

3. Complete Frontend Pages Specification (All 21 Views)

Below is the exhaustive technical breakdown for every single page component in the client/src/pages/ codebase, detailing routes, role access guards, and functional specifications.

Page Component	Route URL	Guard	Functional Specifications
SplashPage.tsx	/	Public	Landing page featuring realistic nurse portraits, brand tagline, and CTAs.
RoleSelectionPage.tsx	/role-selection	Public	Portal for choosing Nurse vs Administrator workspace login.
LoginPage.tsx	/login	Public	JWT authentication screen with quick-fill demo credentials.
NurseDashboard.tsx	/nurse/dashboard	Nurse, Admin	Nurse home portal with active shift banner, quick actions, and schedule preview.
AdminDashboard.tsx	/admin/dashboard	Admin	Executive overview with staffing metrics, ward attendance, and request queues.
NurseRoster.tsx	/nurse/roster	Nurse, Admin	Weekly/monthly roster calendar grid with color-coded shifts & details buttons.
CreateRosterPage.tsx	/admin/roster/create	Admin	Admin schedule builder interface for assigning shifts across hospital wards.
ShiftDetailsPage.tsx	/nurse/shifts/:id	Nurse, Admin	Detailed view of a shift assignment, handover notes, and ward contacts.
CommunicationSimulatorPage.tsx	/nurse/communication-simulator	Nurse, Admin	Scenario selection dashboard for the Gemma 4 AI Patient Simulator.
CommunicationSessionPage.tsx	/nurse/communication-simulator/session/:id	Nurse, Admin	Live roleplay chat room with emotion status pills & real-time Gemma 4 reply.

3. Complete Frontend Pages Specification (Continued)

Page Component	Route URL	Guard	Functional Specifications
CommunicationResultsPage.tsx	/nurse/communication-simulator/session/:id/results	Nurse, Admin	Post-simulation analysis dashboard displaying 8-competency radar scores & advice.
TicTacToeGamePage.tsx	/nurse/break-games/tic-tac-toe	Nurse, Admin	AI Tick & Cross break game with 3 difficulty modes, 10-min timer, and scoreboard.
ShiftSwapPage.tsx	/nurse/shift-swap	Nurse	Form to submit shift exchange requests with target peer nurses.
LeaveRequestPage.tsx	/nurse/leave-request	Nurse	Form to apply for Sick, Annual, or Emergency leave with date selectors.
MyRequestsPage.tsx	/nurse/requests	Nurse	Nurse personal status tracker for pending/approved swaps & leave applications.
PendingRequestsPage.tsx	/admin/requests	Admin	Admin portal to review, approve, or reject pending leave and swap applications.
StaffManagementPage.tsx	/admin/staff	Admin	Staff directory for viewing and updating employee roles & ward departments.
NotificationsPage.tsx	/nurse/notifications	Auth	Real-time push notification center with unread filtering & mark-read actions.
DutyReportPage.tsx	/admin/reports	Admin	Ward staffing analytics and duty hours reporting interface.
ProfilePage.tsx	/nurse/profile	Auth	User profile settings, employee badge ID, and department settings.
PublicPage.tsx	/public	Public	Public ward information landing page.

4. Reusable UI Component Library & Design System Tokens

NurseFlow implements a custom Glassmorphic design system using Tailwind CSS design tokens, HSL purple brand colors (#5142C5), and photorealistic nurse visual assets.

UI Component	File Path	Design Tokens & Functional Properties
Button.tsx	components/ui/Button.tsx	Polymorphic button with primary, secondary, outline, ghost, danger, success variants & spinners.
Card.tsx	components/ui/Card.tsx	Glassmorphic container card with elevation tokens (shadow-nurse-sm, shadow-nurse-md).
Badge.tsx	components/ui/Badge.tsx	Status pill indicators for shift types (Morning, Evening, Night) & approvals (Pending, Approved).
Modal.tsx	components/ui/Modal.tsx	Accessible modal backdrop for shift notes, handover details, and confirmation dialogs.
Input & Select	components/ui/Input.tsx	Custom form input fields and select dropdowns with focus ring states.
StatCard.tsx	components/ui/StatCard.tsx	Dashboard metric display card with icon badges and trend indicators.
RealisticNurseDisplay	components/ui/RealisticNurseDisplay.tsx	Dual photorealistic nurse avatar component displaying female nurse with full hijab & male nurse together.
Sidebar.tsx	components/navigation/Sidebar.tsx	Desktop left navigation menu with active link highlighting & AI feature links positioned above Profile.
Topbar.tsx	components/navigation/Topbar.tsx	Glassmorphic header bar with user avatar, role badge, notification dropdown, and quick logout.
MobileBottomNav.tsx	components/navigation/MobileBottomNav.tsx	Bottom navigation tab bar for mobile viewports.

5. Core Feature Module 1: Smart Duty Roster Calendar Engine

The Roster Calendar module (NurseRoster.tsx) provides nurses and administrators with a spacious, color-coded weekly and monthly duty schedule. Cards are styled to prevent text collision regardless of screen resolution.

Shift Definitions & Color Tokens:

- **Morning Shift:** 7:00 AM – 3:00 PM (Soft Emerald Tint bg-emerald-50/70 border-emerald-200)
- **Evening Shift:** 3:00 PM – 11:00 PM (Soft Amber Tint bg-amber-50/70 border-amber-200)
- **Night Shift:** 11:00 PM – 7:00 AM (Soft Light Purple Tint bg-[#EDE9FE]/50 border-purple-200)
- **Off Duty:** No Duty Assigned (Soft Slate Tint bg-slate-50 border-slate-200)
- **Dedicated Full-Width Button:** Every day card contains a full-width View Details → glass button that opens shift notes and handover instructions without text overlapping.

6. Core Feature Module 2: Gemma 4 AI Patient Communication Simulator

The AI Patient Communication Simulator allows nurses to practice Healthcare Communication in simulated scenarios with realistic patient and caregiver personas. It roleplays realistic situations to develop empathy, active listening, and de-escalation skills.

Pre-Seeded Clinical Training Scenarios:

1. **Anxious Patient:** Pre-surgery fears and anxiety regarding anesthesia risks.
2. **Angry Family Member:** Relatives upset about delayed physician visits in ward 3.
3. **Confused Elderly Patient:** Post-operative disorientation and memory lapse.
4. **Non-Cooperative Patient:** Patient refusing prescribed medication or IV placement.

Dynamic Character Emotion Transition Engine:

The AI engine continuously monitors nurse dialogue. Empathy keywords trigger positive emotional state transitions:

Hostile → Frustrated → Worried → Calm → Reassured

6. Core Feature Module 2: 8-Competency AI Evaluation Radar (Continued)

Upon completing a simulation session, Gemma 4 evaluates the entire conversation transcript and outputs 0–100 scores across 8 key competencies:

Competency Metric	Scoring Criteria & Evaluation Focus
1. Empathy & Compassion	Validating patient feelings & maintaining a caring, supportive tone.
2. Active Listening	Addressing patient questions without skipping medical details.
3. Communication Clarity	Explaining complex medical procedures in clear, understandable terms.
4. Professional Composure	Maintaining calm professionalism under hostile or demanding situations.
5. Emotional Intelligence (EQ)	Recognizing emotional cues and shifting tone appropriately.
6. De-escalation Capability	Neutralizing angry family members or agitated patients effectively.
7. Patient Engagement	Involving the patient actively in care plan decisions.
8. Clinical Confidence	Reassuring the patient with confident healthcare guidance.

7. Core Feature Module 3: Gemma 4 AI Tick & Cross (Tic-Tac-Toe ■■) Break Game

Designed specifically for 10-minute shift breaks, this 100% non-medical relaxation game allows nurses to play Tic-Tac-Toe against Gemma 4 AI to unwind and reset mental focus.

- **3 AI Difficulty Modes:** Easy (Casual Chill), Medium (Smart Rival), and Unbeatable (Gemma 4 Minimax Algorithm).
- **Minimax Algorithm:** Evaluates move trees up to depth 6 for optimal move selection.
- **Scoreboard & Wellness Points:** Tracks Nurse Wins, AI Wins, Draws, and awards Wellness Points to the NurseGameScore model.
- **10-Minute Break Countdown:** Built-in live timer ensures shift breaks stay structured.

8. Core Feature Module 4: Shift Swaps, Leave Portal & Socket.IO Push Alerts

NurseFlow includes administrative tools for duty exchanges, leave management, and instant push notification broadcasting.

- **Shift Swap Requests:** Nurse-to-nurse duty exchange system with target peer selection, reason tracking, and 1-click admin approval portals.
- **Leave Requests Portal:** Annual, Sick, and Emergency leave applications with date range selectors and status badges (Pending, Approved, Rejected).
- **Socket.IO Push Engine:** Bi-directional WebSocket event engine for instant emergency ward broadcasts and unread notification badge syncing.

9. Full Database Schema & Data Dictionary (All 12 Prisma Models)

NurseFlow utilizes **Prisma ORM** with a relational SQLite database (dev.db). Below is the complete technical breakdown of all 12 database models.

1. User Model

Stores staff credentials, role permissions (NURSE, ADMIN, HEAD_NURSE), department assignments, and contact details.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
name	String	Full Name of Staff Member
email	String (Unique)	Login Credential Email
passwordHash	String	Bcrypt Hashed Password
role	String (Default: NURSE)	NURSE ADMIN HEAD_NURSE
employeeId	String (Unique)	Hospital Badge ID
departmentId	String (Optional)	Foreign Key to Department

2. Department Model

Represents hospital ward units (General Ward, ICU, Emergency, Pediatrics).

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
name	String (Unique)	Department Name
description	String (Optional)	Ward Description & Capacity

3. Shift Model

Defines hospital shift timings, color coding, and shift categories.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
name	String	Shift Name (Morning, Evening, Night)
startTime	String	Formatted Start Time (7:00 AM)
endTime	String	Formatted End Time (3:00 PM)
type	String	MORNING EVENING NIGHT OFF

4. Roster Model

Stores daily duty assignments linking nurses, shifts, and departments.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key

nurseld	String	Foreign Key to User (Cascade Delete)
shiftId	String	Foreign Key to Shift (Cascade Delete)
departmentId	String	Foreign Key to Department (Cascade Delete)
date	String	Duty Date (YYYY-MM-DD)
status	String	ON_DUTY SCHEDULED OFF COMPLETED

5. LeaveRequest Model

Tracks leave applications submitted by nurses and review decisions.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
nurseld	String	Foreign Key to User (Applicant)
leaveType	String	Annual Leave Sick Leave Emergency
fromDate	String	Start Date (YYYY-MM-DD)
toDate	String	End Date (YYYY-MM-DD)
status	String	PENDING APPROVED REJECTED

6. ShiftSwapRequest Model

Manages nurse-to-nurse shift exchange applications.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
requesterId	String	Foreign Key to Requesting Nurse
targetNurseld	String	Foreign Key to Target Peer Nurse
originalShiftId	String	Foreign Key to Original Shift
status	String	PENDING APPROVED REJECTED

9. Full Database Schema & Data Dictionary (Continued)

7. Notification Model

Stores user-specific push notifications and emergency ward alerts.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
userId	String	Foreign Key to User
title	String	Notification Header Title
message	String	Push Notification Content
type	String	SHIFTS LEAVE SWAP ROSTER ALERT INFO
isRead	Boolean	Read/Unread State Flag

8. CommunicationScenario Model

Defines AI simulation roleplay scenarios and character personalities.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
title	String	Scenario Header Title
category	String	Patient Family Elderly Emergency
description	String	Detailed Context & Medical Background
characterRole	String	Patient Family Member Elderly Patient
personality	String	Anxious Hostile Confused Demanding

9. CommunicationSession Model

Tracks active and completed AI roleplay sessions.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
userId	String	Foreign Key to User (Nurse)
scenarioId	String	Foreign Key to CommunicationScenario
characterRole	String	Roleplayed Character Persona
status	String	ACTIVE COMPLETED ABANDONED

10. CommunicationMessage Model

Stores individual dialogue turns and emotion states in roleplay chats.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key

sessionId	String	Foreign Key to CommunicationSession
role	String	NURSE or PATIENT
content	String	Transcribed Dialogue Text
emotion	String	Calm Worried Confused Frustrated Angry Reassured

11. CommunicationAnalysis Model

Stores 8-competency scores and AI clinical feedback reports.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
sessionId	String (Unique)	Foreign Key to CommunicationSession
overallScore	Int	Calculated Overall Score (0-100)
empathyScore	Int	Empathy & Compassion Score
activeListeningScore	Int	Active Listening Score
clarityScore	Int	Communication Clarity Score
feedback	String	Comprehensive AI Coaching Summary Text

12. NurseGameScore Model

Tracks nurse wins, AI wins, draws, and earned wellness points in Tick & Cross game.

Field Name	Data Type & Constraints	Description
id	String (UUID)	Primary Key
userId	String	Foreign Key to User (Nurse)
gameType	String	Break Game Category
nurseWins	Int	Total Nurse Victory Count
aiWins	Int	Total Gemma 4 AI Victory Count
pointsEarned	Int	Total Shift Wellness Points Earned

10. Full Backend REST API Endpoints Specification

All authenticated endpoints require an Authorization: Bearer <token> HTTP header.

Method	Endpoint Route	Functionality Description	Access
POST	/api/auth/login	Authenticate user credentials & return JWT token	Public
POST	/api/auth/register	Register new nurse or administrator account	Public
GET	/api/users	Fetch list of all staff nurses & employees	Admin
GET	/api/rosters	Fetch duty rosters (filter by date/nurse)	Auth
POST	/api/rosters	Assign shift roster entry to nurse	Admin
POST	/api/leave-requests	Submit leave application (Sick, Annual)	Nurse
POST	/api/shift-swaps	Request shift exchange with target nurse	Nurse
POST	/api/notifications/send-alert	Broadcast ward emergency push notification	Admin
GET	/api/communication-simulator/scenarios	Fetch roleplay scenarios list	Auth
POST	/api/communication-simulator/sessions	Initialize new AI roleplay session	Auth
POST	/api/communication-simulator/sessions/:id/messages	Send nurse dialogue & get AI roleplay reply	Auth
POST	/api/communication-simulator/sessions/:id/end	End session & generate 8-competency evaluation	Auth
POST	/api/games/tic-tac-toe/move	Compute Gemma 4 AI move (Minimax)	Auth
POST	/api/games/tic-tac-toe/score	Save match outcome & update wellness points	Auth

11. Security, Role Guards & Operational Setup

Security & Authorization Guards:

- **Authentication Token Middleware:** Express middleware `authenticateToken` verifies JWT validity.
- **Role-Based Protected Routes:** React Router `ProtectedRoute` component checks role permissions (NURSE, ADMIN, HEAD_NURSE).

Local Setup & Deployment Commands:

```
# 1. Install Server & Client Dependencies
cd server && npm install
cd ../client && npm install

# 2. Sync Database Schema & Run Seed Data
cd ../server
npx prisma db push
npx ts-node prisma/seed.ts

# 3. Start Development Servers
npm run dev    # inside server (Port 5000)
npm run dev    # inside client (Port 5173)
```

NurseFlow Complete System Manual | GitHub: [Sagheer1122/Arbisoft-GDG-Hackathon](#)